Basic Graph Theory

Béla Bollobás
University of Cambridge

Robert Morris
IMPA, Rio de Janeiro

Contents

Introduction

Graphs are one of the simplest mathematical objects to define – they are just a set of pairs ('edges') of elements of some underlying vertex set – but they nevertheless display remarkable complexity. Their study has exploded in popularity in recent decades due to the fundamental role that they play in computer science and the emerging areas of network science and artificial intelligence, as well as in more traditional scientific fields such as statistical physics and biology, in which the complexity of the system emerges naturally from relatively simple local interactions. At the same time, the area of graph theory has matured into a central field of pure mathematics, with close connections to number theory, analysis, probability theory, algebra, and high-dimensional geometry.

There is another reason, however, for the popularity of graph theory amongst both students and researchers: it is perhaps the most accessible way of experiencing true mathematical beauty. As the reader will see, graph theory is overflowing with questions that are easy to understand, surprisingly difficult to solve, and which are finally unlocked by a simple and beautiful idea that can be applied in many other places. One of the aims of this book is to arm the student with a collection of tools of this kind: insights, such as the probabilistic method, that transform apparently insurmountable problems into mere exercises.

This leads us to an important question: who is this book written for? We have attempted to write it with essentially no prerequisites, and it could therefore (in theory) be understood by a talented high-school student. In practice, however, over 50 years of teaching experience has taught us that the study of graph theory requires some mathematical 'maturity' (whatever that means), and we therefore recommend that the book be used for a final-year undergraduate or Master's level course. More precisely, Chapters 1–4 and the first half of Chapter 5 should be accessible to final-year undergraduate students, while the second half of Chapter 5 covers some slightly more advanced material.

At the end of each chapter we have included 30–40 exercises, some of which are only intended to check that the reader has understood the material, while others require (sometimes significant) additional ideas to solve. We encourage the reader to attempt all of the exercises, although solving them all would be a real achievement. We note that many of the exercises from the earlier chapters are used in later chapters, and several of the exercises in the later chapters hint at problems and techniques that are too advanced to be included in this book, but that the reader would be likely to meet in a graduate-level course on graph theory. We have also included, at the end of each chapter, a few historical remarks and suggestions of further reading for those students who would like to dive more deeply into the topics covered in the chapter.

Finally, we would like to emphasize that although several of the theorems discussed in the book are more than 100 years old, graph theory is still a young and rapidly evolving subject, with numerous significant breakthroughs occurring in just the past few years. It is also an area that, as mentioned above, is overflowing with beautiful open problems, and it is these problems (rather than a desire to build machinery for its own sake) that tend to drive the subject forward, the theory emerging out of attacks on these problems, rather than the other way around. This emphasis on open problems, and on beauty, makes the study of graph theory an especially joyful enterprise, and we hope that this book gives the reader a taste of how much fun it is to work in the area.

B. B.
Cambridge

R. M.
Rio de Janeiro

March 26th, 2025

1
Basic Graph Theory

In this chapter we will begin gently, by introducing the reader to a few basic types of graphs: trees, bipartite graphs, triangle-free graphs, and to some simple parameters of a graph, including the independence number and the chromatic number. Our aim is to warm ourselves up for the later chapters by getting used to thinking about graphs, and to proving things about them. We encourage the reader to try to prove each of the results in this chapter herself, before reading the proofs.

1.1 First steps

The aim of this first section is to introduce a few basic concepts that will be used throughout the book, and to illustrate these concepts with some simple (but surprisingly useful) facts. To get started, let us define the central object of study in this book: the graph.

Definition 1.1.1. A graph G consists of a set $V(G)$ of *vertices* and a set $E(G)$ of *edges*, where each edge is a pair of vertices $\{u,v\} \subset V(G)$.

We denote by $v(G)$ and $e(G)$ the number of vertices and edges of a graph G, respectively. We shall also write uv instead of $\{u,v\}$ when referring to an edge, and say that u and v are *incident to* the edge uv. For example, the *complete graph* K_n has n vertices and edge set

$$E(K_n) = \big\{ uv : u,v \in V(K_n) \text{ and } u \neq v \big\},$$

and therefore $v(K_n) = n$ and $e(K_n) = \binom{n}{2}$. We also write

$$N_G(v) = \big\{ u \in V(G) : uv \in E(G) \big\} \qquad \text{and} \qquad d_G(v) = |N_G(v)|$$

for the *neighbourhood* and the *degree* of a vertex v in G.

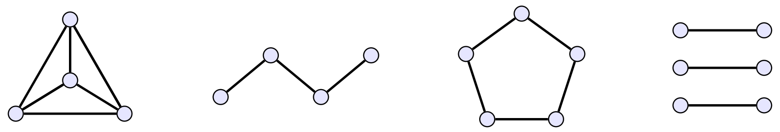

Figure 1.1 Four graphs: K_4 (the complete graph with four vertices), P_3 (the path of length 3), C_5 (the cycle of length 5), and a matching of size 3.

We are now ready for our first result about graphs. The following simple but fundamental fact is known as the 'handshaking lemma'.

Lemma 1.1.2. *For any graph G,*

$$\sum_{v \in V(G)} d_G(v) = 2e(G). \tag{1.1}$$

Proof We count the pairs (v, e), where $v \in V(G)$ and $e \in E(G)$ is an edge incident to v, in two different ways. First, recall that a vertex v is incident to exactly $d_G(v)$ edges of G, and therefore the number of such pairs (v, e) is equal to the sum on the left-hand side of (1.1). On the other hand, each edge is counted exactly twice (once for each of its vertices), so there are exactly $2e(G)$ such pairs, as required. □

The proof above is a simple example of *double counting* (which just means counting the same thing in two different ways), a technique that we shall encounter numerous times during this book.

1.1.1 Subgraphs, paths, and cycles

In graph theory we are often interested in when one graph 'contains' another one. The simplest (and most important) notion of containment is that of being a *subgraph*: we say that H is a subgraph of G if

$$V(H) \subset V(G) \qquad \text{and} \qquad E(H) \subset E(G),$$

and write $H \subset G$ to denote the fact that H is a subgraph of G. We will also say that G *contains a copy* of H if there exists a subgraph $H' \subset G$ that is isomorphic[1] to H. For example, if $u, v, w \in V(G)$ and $uv, uw, vw \in E(G)$, then G contains a copy of the *triangle* K_3.

[1] We say that two graphs G and H are isomorphic if there is a bijection between the vertices of G and H that maps every edge to an edge, and every non-edge to a non-edge. We will write $G = H$ to mean that G is isomorphic to H.

Note that there are two natural ways to generalise the triangle K_3: to the complete graph K_k on k vertices, and to the *cycle* C_k of length $k \geqslant 3$, which has vertex set $\{v_1, \dots, v_k\}$ and edge set

$$E(C_k) = \big\{v_1v_2, v_2v_3, \dots, v_{k-1}v_k, v_kv_1\big\}.$$

We will sometimes write $C = (v_1, \dots, v_k, v_1)$ to denote the cycle with this edge set. The complete graph K_4 and the cycle of length 5 are illustrated in Figure 1.1.

Similarly, we define P_k, the *path* of length k, to be the graph with vertex set $\{v_0, \dots, v_k\}$ and edge set

$$E(P_k) = \big\{v_0v_1, v_1v_2, \dots, v_{k-1}v_k\big\}.$$

We will sometimes write $P = (v_0, \dots, v_k)$ to denote the path with this edge set, and refer to v_0 and v_k as the *end-vertices* of P. The path P_3 is illustrated in Figure 1.1. We will meet the fourth graph in the figure, which is called a *matching*, in Section 1.1.2.

In order to state our second result, let us write

$$\delta(G) = \min\big\{d_G(v) : v \in V(G)\big\}$$

to denote the *minimum degree* of a graph G.

Lemma 1.1.3. *If $\delta(G) \geqslant 2$, then G contains a cycle.*

Proof Let $P = (v_0, \dots, v_k)$ be a maximal path in the graph G. Since $\delta(G) \geqslant 2$, we have $N_G(v_0) \neq \{v_1\}$, and so there must exist an edge $uv_0 \in E(G)$ with $u \neq v_1$. Now, if $u \notin V(P)$, then P can be extended to a longer path by adding the edge uv_0, contradicting our assumption that P is maximal. It follows that $u = v_i$ for some $i \in \{2, \dots, k\}$, and therefore $(v_0, v_1, \cdots, v_i, v_0)$ is a cycle in G. □

What if our graph G does not have minimum degree 2, but has many edges? Can we still find a cycle? By the handshaking lemma, we know that if $e(G) \geqslant v(G)$, then the *average degree* of G satisfies

$$\frac{1}{v(G)} \sum_{v \in V(G)} d_G(v) = \frac{2e(G)}{v(G)} \geqslant 2,$$

but this is not enough to apply Lemma 1.1.3. However, the following lemma implies that in this case G contains a *subgraph* with minimum degree 2, and we can then apply the lemma to that subgraph instead. More generally, it says that every graph G has a subgraph whose minimum degree is greater than half the average degree of G.

Lemma 1.1.4. *If the average degree of a graph G is at least $2d$, then G contains a subgraph with minimum degree strictly greater than d.*

Proof There is really only one thing we can do: if a vertex has degree at most d, then we must remove it from the graph. Doing so repeatedly, we either find a subgraph $H \subset G$ with $\delta(H) > d$, as required, or we obtain an ordering $v_1, \ldots, v_n$ of the vertices of G such that

$$\big| N_G(v_i) \cap \big\{ v_{i+1}, \ldots, v_n \big\} \big| \leqslant d \tag{1.2}$$

for every $i \in \{1, \ldots, n\}$. But if (1.2) holds, then

$$e(G) = \sum_{i=1}^{n-1} \big| N_G(v_i) \cap \big\{ v_{i+1}, \ldots, v_n \big\} \big| < dn,$$

and hence, by the handshaking lemma, the average degree of G satisfies

$$2d \leqslant \frac{1}{n} \sum_{v \in V(G)} d_G(v) = \frac{2e(G)}{n} < 2d,$$

which is a contradiction. It follows that there does not exist an ordering of the vertices of G satisfying (1.2), and therefore G contains a subgraph with minimum degree greater than d, as claimed. □

Observe that the average degree of a path converges to 2 as the length of the path tends to infinity, but a path contains no subgraph H with $\delta(H) > 1$, so the assumption in Lemma 1.1.4 cannot be weakened.

Combining Lemmas 1.1.3 and 1.1.4, we obtain our first theorem.

Theorem 1.1.5. *If $e(G) \geqslant v(G)$, then G contains a cycle.*

Proof By the handshaking lemma, G has average degree at least 2. Therefore, by Lemma 1.1.4, there exists a subgraph $H \subset G$ with minimum degree at least 2. By Lemma 1.1.3, it follows that there exists a cycle C in H, and since $H \subset G$, we also have $C \subset G$. □

In other words, Theorem 1.1.5 says that if a graph G with n vertices does not contain a cycle, then it has at most $n-1$ edges. Note that this bound is best possible, since the path of length $n-1$ has n vertices, $n-1$ edges, and does not contain a cycle. We will study *extremal* problems of this type (how many edges do we need to guarantee the existence of a certain substructure in G) in much more detail in Chapter 2.

1.1.2 Induced subgraphs

A more restrictive way in which one graph can 'contain' another is as an *induced subgraph*. Given a graph G and a set of vertices $U \subset V(G)$, we will write $G[U]$ to denote the subgraph of G *induced by* U, that is, the graph with vertex set U and edge set

$$E\big(G[U]\big) = \big\{uv \in E(G) : u, v \in U\big\}.$$

If $H = G[U]$ for some $U \subset V(G)$, then we write $H \leqslant G$ and say that H is an *induced subgraph* of G, or that G *contains an induced copy* of H. For example, if $uv, vw \in E(G)$ but $uw \notin E(G)$, then the graph G contains an induced copy of P_2, the path of length 2.

Given graphs G and H, we will say that G is *H-free* if it does not contain a copy of H, and similarly that G is *induced-H-free* if it does not contain an induced copy of H. To illustrate these two definitions, let's consider a very simple special case: when $H = P_2$. We will write

$$\Delta(G) = \max\big\{d_G(v) : v \in V(G)\big\} \tag{1.3}$$

for the *maximum degree* of a graph G, and say that a graph G is a *matching* if $\Delta(G) \leqslant 1$ (that is, every vertex of G has degree at most 1). The next observation follows immediately from the definitions.

Observation 1.1.6. *Every P_2-free graph is a matching.*

Proof If G contains a vertex of degree at least 2, then let $u, w \in N_G(v)$ and observe that the graph H with vertex set $\{u, v, w\}$ and edge set $\{uv, vw\}$ is a subgraph of G and also a copy of P_2. $\square$

Note that a matching consists of a collection of disjoint edges and some 'isolated' vertices (that is, vertices of degree zero).

The family of graphs that do not contain an induced copy of P_2 is a little larger. A *clique* is a copy of K_k for some $k \in \mathbb{N}$.

Theorem 1.1.7. *Every induced-P_2-free graph is formed by taking the union of a collection of vertex-disjoint cliques.*

Proof We will prove the claim by induction on the number of vertices. Note that the base case is trivial, since the unique graph with $v(G) = 1$ is a clique. Let G be an induced-P_2-free graph, and assume that the claim holds for all graphs with at most $v(G) - 1$ vertices.

Let $A \subset V(G)$ be the vertex set of a maximal clique in G, and let $B = V(G) \setminus A$. Note that $G[B]$, the subgraph of G induced by B, is induced-P_2-free, since any induced copy of P_2 in $G[B]$ would also be an induced copy of P_2 in G. By the induction hypothesis, it follows that $G[B]$ is the union of a collection of vertex-disjoint cliques. It will therefore suffice to show that there are no edges of G between A and B.

To prove this, suppose that $ab \in E(G)$ with $a \in A$ and $b \in B$. Since A is maximal, there must exist a vertex $c \in A$ such that $bc \notin E(G)$. Now, since A is a clique, we have $ac \in E(G)$, and therefore the set $\{a, b, c\}$ induces a copy of P_2 in G, which contradicts our assumption that G is induced-P_2-free. It follows that there are no edges between A and B and, as noted above, this completes the proof. $\square$

The following alternative proof of Theorem 1.1.7 is shorter, and also motivates the topic of the next subsection.

Second proof of Theorem 1.1.7 We claim that if there is a path in G between two (distinct) vertices u and v, then $uv \in E(G)$. To see this, let $P = (w_1, \ldots, w_k)$ be a path of minimum length between u and v (so $w_1 = u$ and $w_k = v$). If $k \geqslant 3$, then the set $\{w_1, w_2, w_3\}$ induces a copy of P_2 in G, since $w_1w_3 \notin E(G)$ by the minimality of P.

As we shall see in Section 1.1.3, below, writing $u \sim v$ if there exists a path in G between u and v defines an equivalence relation, and our claim implies that each equivalence class is a clique, as required. $\square$

Note that matchings are P_2-free, and that the union of a collection of disjoint cliques is induced-P_2-free, so in fact Observation 1.1.6 and Theorem 1.1.7 characterise P_2-free and induced-P_2-free graphs.

1.1.3 Connected graphs

A fundamental notion in graph theory is that of being *connected*.

Definition 1.1.8. We say that a graph G is *connected* if for every pair of vertices $u, v \in V(G)$, there exists a path between u and v in G.

The following theorem provides a useful alternative definition.

Theorem 1.1.9. *A graph G is connected if and only if there is an edge between A and $V(G) \setminus A$ for every set $A \subset V(G)$ with $0 < |A| < v(G)$.*

Proof Suppose first that there exists a set $A \subset V(G)$ with no edges between A and $B = V(G) \setminus A$, and that $u \in A$ and $v \in B$. We claim that there is no path in G between u and v. Indeed, if $(w_1, \ldots, w_k)$ were a path in G with $w_1 = u$ and $w_k = v$, then, letting i be minimal such that $w_i \in B$, we see that $w_{i-1}w_i$ is an edge of G with $w_{i-1} \in A$ and $w_i \in B$. This contradiction shows that G is not connected, as claimed.

On the other hand, suppose that G is not connected, and fix a pair of vertices $u, v \in V(G)$ with no path between u and v in G. Define $A \subset V(G)$ to be the set of vertices of G that can be reached by a path starting at u, and let $B = V(G) \setminus A$. We claim that

(i) A and B are both non-empty, and

(ii) there are no edges of G between A and B,

as required. For (i), simply note that $u \in A$ and $v \in B$. To prove (ii), note that if $ab \in E(G)$ with $a \in A$ and $b \in B$, then adding the edge ab to the path from u to a would produce a path in G between u and b. □

Theorem 1.1.9 has the following simple but interesting consequence. Given a graph G, we write G^c for the *complement* of G; that is, the graph with vertex set $V(G)$ and edge set $\binom{V(G)}{2} \setminus E(G)$, where we write $\binom{A}{2}$ to denote the collection of subsets of A of size 2.

Corollary 1.1.10. *For every graph G, either G or G^c is connected.*

Proof Suppose that G is not connected. By Theorem 1.1.9, there exists a partition $V(G) = A \cup B$ into non-empty sets such that $ab \notin E(G)$ (and therefore $ab \in E(G^c)$) for every $a \in A$ and $b \in B$. It follows that every pair of vertices of G^c is connected either by an edge, or by a path of length two, and hence G^c is connected, as required. □

In the proof of Theorem 1.1.9 we defined the set of vertices that can be reached by a path from a given vertex u. We call the subgraph of G that is induced by this set of vertices the *component* of u in G.

We claim that the component of a vertex is connected: if there exist paths from u to x and from u to y in a graph G, then there exists a path between x and y in G. This is an immediate consequence of the following simple fact. We call a sequence of edges $v_1v_2, v_2v_3, \ldots, v_{k-1}v_k$ in a graph G (where vertices can occur more than once) a *walk* in G.

Observation 1.1.11. *Two vertices in a graph G are connected by a walk in G if and only if they are connected by a path in G.*

Proof Every path is a walk, so it will suffice to show that if two vertices u and w are connected by a walk in a graph G, then they are also connected by a path. To do so, simply consider a minimal walk $uv_1, v_1v_2, \ldots, v_kw$ between u and w. No vertex can be used more than once, since then we could remove the edges between two appearances of that vertex and construct a shorter walk between u and w. $\square$

We can now easily deduce the following important fact.

Lemma 1.1.12. *Let G be a graph and $u \in V(G)$. The component of u in G is the unique maximal connected subgraph of G containing u.*

Proof Let A be the set of vertices that can be reached by a path in G from u. To see that $G[A]$ is connected, let $v, w \in A$ and let P and Q be paths in G from u to v and w respectively. Following P backwards and then Q forwards gives a walk from v to w in G, and therefore, by Observation 1.1.11, there exists a path from v to w.

Now, simply note that, by Definition 1.1.8, no connected subgraph of G can contain both u and a vertex $v \in V(G) \setminus A$, since there is no path in G between u and v. It follows that every connected subgraph of G containing u is a subgraph of $G[A]$, as claimed. $\square$

We will usually refer to the components of the vertices of a graph G as the *connected components* of G. Note that G has a single connected component if and only if it is connected. We also have the following simple fact, which we used in the second proof of Theorem 1.1.7.

Lemma 1.1.13. *The connected components of a graph G partition both the vertices and the edges of G.*

Proof By Lemma 1.1.12, each vertex is in a unique connected component of G. Moreover, if $uv \in E(G)$ then u and v are in the same connected component, and hence uv is an edge of that component. $\square$

We can also now restate Theorem 1.1.7 as follows: each connected component of an induced-P_2-free graph is a clique.

1.2 Trees

Trees are among the simplest and most widely used families of graphs, with a vast number of applications across many different scientific fields, including physics, biology, theoretical computer science, data science, network science and artificial intelligence. They also play a fundamental role within graph theory itself, and although in this short section we will only prove a few basic properties of trees, we will apply these results numerous times in later chapters. Proving these simple results will also give us an opportunity to introduce several useful proof techniques.

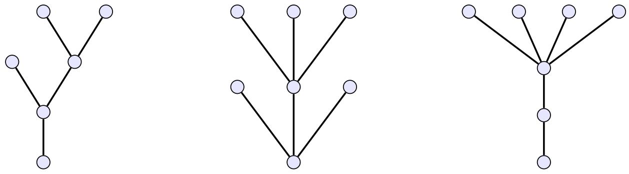

Figure 1.2 A graph in which each connected component is a tree.

We begin by defining what it means for a graph to be a tree.

Definition 1.2.1. A *tree* is a connected graph that contains no cycles.

For example, note that paths are trees. We will sometimes say that a graph is *acyclic* to mean that it does not contain any cycles. Our first fact about trees is that the number of edges of a tree is determined by the number of vertices.

Theorem 1.2.2. *Every tree with n vertices has exactly $n-1$ edges.*

We will prove the theorem by induction on n. In order to do so, we will first need to make two simple observations about trees. A vertex v of a graph G is called a *leaf* if it has exactly one neighbour in G.

Lemma 1.2.3. *Every tree with at least two vertices has a leaf.*

Proof Let G be a tree, let P be a maximal path in G, and let v be one of the end-vertices of P. Since P is maximal, for every edge $uv \in E(G)$ we must have $u \in V(P)$. However, if v had two neighbours in P, then G would contain a cycle. Since G is a tree, and therefore acyclic, and has at least two vertices, it follows that v must be a leaf of G. $\square$

We will combine Lemma 1.2.3 with the following observation, which will allow us to use induction. Let us write $G - v$ for the subgraph of G obtained by removing the vertex v, and all edges incident to v.

Lemma 1.2.4. *Let v be a leaf of a tree G. Then $G - v$ is also a tree.*

Proof Note first that $G - v$ does not contain a cycle, since G is acyclic, and removing a vertex cannot create a cycle. To prove that $G - v$ is connected, let $u, w \in V(G) \setminus \{v\}$ be vertices of $G - v$, and let P be a path between u and w in G. Since v is a leaf, P cannot pass through v (since a path passes through each vertex at most once), and it is therefore also a path between u and w in $G - v$. Since u and w were arbitrary vertices of $G - v$, it follows that $G - v$ is connected, as required. □

We can now easily deduce that trees have $n - 1$ edges.

Proof of Theorem 1.2.2 The proof is by induction on n. Note first that the lemma is true when $n = 1$, because the only graph with one vertex is K_1, which has zero edges. So let G be a tree with $n \geqslant 2$ vertices, and assume that $e(G) = n - 2$ for every tree with $n - 1$ vertices.

By Lemma 1.2.3, the tree G has a leaf. Let v be a leaf of G and observe that, by Lemma 1.2.4, the graph $G - v$ is a tree with $n - 1$ vertices. Therefore, by the induction hypothesis, $G - v$ has exactly $n - 2$ edges. Since $d_G(v) = 1$, it follows that

$$e(G) = e(G - v) + 1 = n - 1,$$

as required. □

Lemmas 1.2.3 and 1.2.4 can be used to give simple inductive proofs of numerous properties of trees. For example, recall that $\delta(G)$ denotes the minimum degree of a graph G and that we showed in Lemma 1.1.3 that every graph with $\delta(G) \geqslant 2$ contains a cycle. The following theorem gives a sharp bound on the minimum degree required to guarantee that a graph contains *every* tree with a given number of vertices.

Theorem 1.2.5. *Let G be a graph with $\delta(G) \geqslant k$, and let T be a tree with $k + 1$ vertices. Then $T \subset G$.*

Proof The proof is by induction on k. Note that the claim holds when $k = 1$, since any graph with at least one edge contains the unique tree with two vertices. So let T be a tree with $k + 1 \geqslant 3$ vertices and assume that the theorem is true for all trees with k vertices.

By Lemmas 1.2.3 and 1.2.4, there exists a leaf v of T, and the graph $T' = T - v$ is a tree with k vertices. Since $\delta(G) \geqslant k - 1$, it follows from the induction hypothesis that T' is a subgraph of G. Let $A \subset V(G)$ be the vertex set of a copy of T' in G, and let $u \in A$ be the vertex corresponding to the unique neighbour of v in the tree T.

Now, note that $d_G(u) \geqslant \delta(G) \geqslant k$. Since T' has k vertices, it follows that u has a neighbour in $V(G) \setminus A$. Adding this neighbour to T', we obtain a copy of T in G, as required. □

To see that the bound in Theorem 1.2.5 is best possible, consider the graph in which each component is a copy of K_k. This graph has minimum degree $k - 1$ and does not contain any tree with $k + 1$ vertices.

One of the reasons that trees are useful is that they can easily be found as subgraphs. For example, Theorem 1.2.5 allows us to find any specific tree in a graph with sufficiently high minimum degree. If we are content with finding *any* (large) tree, however, then life is even easier. We say that a tree T is a *spanning tree* of a graph G if it is a subgraph of G and has the same set of vertices as G.

Theorem 1.2.6. *Every connected graph has a spanning tree.*

We will deduce Theorem 1.2.6 from the following fact about graphs that are connected but not acyclic. If G is a graph and $e \in E(G)$, then we write $G - e$ to denote the graph obtained from G by removing the edge e; that is, the graph with vertex set $V(G)$ and edge set $E(G) \setminus \{e\}$.

Lemma 1.2.7. *Let G be a graph, let C be a cycle in G, and let e be an edge of C. If G is connected, then $G - e$ is also connected.*

Proof The idea is simply that any path that uses the edge e can be 'rerouted' around the remainder of the cycle. Indeed, let $u, v \in V(G)$, and let P be a path in G between u and v. If P does not use the edge e, then $P \subset G - e$, so assume that $e \in E(P)$. Now, replacing e by the path $C - e$ gives a walk in $G - e$ between u and v, and therefore, by Observation 1.1.11, there exists a path in $G - e$ from u to v. Since u and v were arbitrary vertices of G, it follows that $G - e$ is connected. □

Proof of Theorem 1.2.6 Suppose that G is a minimal counterexample to the theorem. Then G is connected, but is not itself a tree, and hence must contain a cycle C. Choose any edge $e \in E(C)$ and observe that $G - e$ is connected, by Lemma 1.2.7. By the minimality of G, it follows that $G - e$ has a spanning tree T, and since $V(T) = V(G)$, it follows that T is also a spanning tree of G. □

Before continuing, let's quickly note that each of the properties in Definition 1.2.1 provides one of the bounds in Theorem 1.2.2.

Corollary 1.2.8. *Let G be a graph with n vertices.*

(a) If G is acyclic, then $e(G) \leqslant n - 1$.
(b) If G is connected, then $e(G) \geqslant n - 1$.

Proof Part (a) follows immediately from Theorem 1.1.5, which said that if $e(G) \geqslant n$, then G contains a cycle. For part (b), observe that G contains a spanning tree T, by Theorem 1.2.6, and that T has $n - 1$ edges, by Theorem 1.2.2. Hence $e(G) \geqslant e(T) = n - 1$, as required. □

It turns out that there are several different ways to characterise the family of trees. The following alternative definition is often useful.

Theorem 1.2.9. *A graph G is a tree if and only if there is exactly one path between any two vertices of G.*

Proof Observe first that G is connected if and only if there is at least one path between any two vertices of G, by Definition 1.1.8. It will therefore suffice to show that G contains a cycle if and only if there are two paths between some pair of vertices $u, v \in V(G)$.

If G contains a cycle C, then clearly there are two vertices of C with at least two paths between them. So let P and Q be distinct paths in G between u and v, and let H be the graph with vertex set $V(P) \cup V(Q)$ and edge set $E(P) \cup E(Q)$. We claim that $e(H) \geqslant v(H)$. To see this, note that $e(P) = v(P) - 1$, and follow the path Q from u to v, noting that whenever we arrive at a vertex not in $V(P)$, we use an edge that is not in $E(P)$. Since there must be at least one step in which we arrive at a vertex in $V(P)$ using an edge not in $E(P)$, the claim follows.

Now, since $e(H) \geqslant v(H)$, it follows by Theorem 1.1.5 that $H \subset G$ contains a cycle, as required. □

The following theorem gives four further equivalent ways in which one can define the family of trees.[2]

Theorem 1.2.10. *Let G be a graph with n vertices. Then*

$$\begin{aligned} G \text{ is a tree} \quad &\Leftrightarrow \quad G \text{ is a minimal connected graph} \\ &\Leftrightarrow \quad G \text{ is a maximal acyclic graph} \\ &\Leftrightarrow \quad e(G) = n - 1 \text{ and } G \text{ is connected} \\ &\Leftrightarrow \quad e(G) = n - 1 \text{ and } G \text{ is acyclic.} \end{aligned}$$

The proof of Theorem 1.2.10 is not difficult and is a useful exercise in applying the definitions and results of this section; we therefore leave the details to the reader. Some further properties of trees (and collections of trees, which are called 'forests') are given in Exercises 1.1–1.5.

Finally, let us state one more famous fact about trees, which was first proved by Borchardt in 1860, but is known as Cayley's formula.

Theorem 1.2.11. *There are n^{n-2} trees with vertex set $\{1, \ldots, n\}$.*

A bijective proof of this theorem is described in Exercise 1.42, though we warn the reader that filling in the details takes some thought!

[2] Here, and throughout the book, we write $\Leftrightarrow$ to mean "if and only if".

1.3 Bipartite graphs

An *independent set* in a graph G is a set of vertices $A \subset V(G)$ that contains no edges of G. A graph G is *bipartite* if its vertex set can be partitioned into two independent sets A and B. We call such a pair (A, B) a *bipartition* of G, and A and B the *parts* of the bipartition.

To get warmed up, let us begin by observing that trees are bipartite.

Theorem 1.3.1. *Every tree is bipartite.*

In order to prove this theorem, we will introduce the following useful definition: the *distance* between two vertices u and v in a connected graph G, denoted $d(u, v)$, is the length of the shortest path between u and v in G. In particular, recalling Theorem 1.2.9, if G is a tree, then $d(u, v)$ is the length of the unique path between u and v.

Proof of Theorem 1.3.1 Let G be a tree, fix a vertex $u \in V(G)$, and define

$$A = \big\{ v \in V(G) : \, d(u, v) \text{ is even} \big\} \qquad \text{and} \qquad B = V(G) \setminus A.$$

That is, A is the set of vertices of G whose distance from u is even, and B is the set of vertices of G whose distance from u is odd.

We claim that A and B are both independent sets of G. To see this, suppose that vw is an edge of G, and note that $|d(u, v) - d(u, w)| \leqslant 1$. Moreover, if $d(u, v) = d(u, w)$, then the unique path P from u to v does not pass through w, and therefore adding the edge vw to P would give a different path between u and w, contradicting Theorem 1.2.9. It follows that $d(u, v)$ and $d(u, w)$ do not have the same parity, and hence A and B are independent sets, as claimed. □

Bipartite graphs, however, unlike trees, do not need to be acyclic (or connected). In particular, every *even cycle* C_{2k} is bipartite, since if

$$V(C_{2k}) = \big\{ v_1, \ldots, v_{2k} \big\} \quad \text{and} \quad E(C_{2k}) = \big\{ v_1 v_2, v_2 v_3, \ldots, v_{2k} v_1 \big\},$$

then the vertex set of C_{2k} can be partitioned into the independent sets

$$A = \big\{ v_1, v_3, \ldots, v_{2k-1} \big\} \qquad \text{and} \qquad B = \big\{ v_2, v_4, \ldots, v_{2k} \big\}.$$

On the other hand, as readers can easily check for themselves, no *odd cycle* C_{2k+1} is bipartite. It follows that no bipartite graph contains an odd cycle, since every subgraph of a bipartite graph is also bipartite.

The following fundamental theorem, which was first discovered by Kőnig in 1936, shows that containing an odd cycle is in fact the only obstacle to a graph being bipartite.

Theorem 1.3.2. *Let G be a graph. Then*

$$G \text{ is bipartite} \quad \Leftrightarrow \quad G \text{ contains no odd cycle.}$$

As noted above, one direction of the equivalence follows from the fact that odd cycles are not bipartite, and so any graph containing an odd cycle is also not bipartite. We therefore only need to show that if G contains no odd cycle, then it is bipartite.

We will give two different proofs of this fact; in the first we will use Theorems 1.2.6 and 1.3.1.

First proof of Theorem 1.3.2 Let G be a minimal counterexample to the theorem, so G contains no odd cycle but is not bipartite. Suppose first that G is not connected. By the minimality of G, each connected component of G is bipartite, and therefore G is also bipartite.

We may therefore assume that G is connected. By Theorem 1.2.6, it follows that G has a spanning tree T, and by Theorem 1.3.1, the tree T is bipartite, so we may fix a bipartition (A, B) of T. We claim that A and B are independent sets in G.

To see this, let $uv \in E(G)$, and suppose that u and v are both in the same part of the bipartition (A, B). Since T is connected, there exists a path P in T between u and v; moreover, the length of P is even, since A and B are independent in T, and therefore each edge of P has one end-vertex in each of A and B. It follows that adding the edge uv to P gives an odd cycle in G, a contradiction. This implies that A and B are independent sets in G, and hence that G is bipartite, as required. $\square$

For the second proof of Theorem 1.3.2, we need to introduce another simple but important concept: a *cut-vertex* of a connected graph G is a vertex $v \in V(G)$ for which the graph $G - v$ is disconnected.

Lemma 1.3.3. *If G is a connected graph, then there exists a vertex $v \in V(G)$ that is not a cut-vertex of G.*

Proof By Theorem 1.2.6, the graph G has a spanning tree T, and by Lemma 1.2.3, the tree T has a leaf v. Since $T - v$ is contained in $G - v$, and is a tree, by Lemma 1.2.4, it follows that $G - v$ is connected, and therefore v is not a cut-vertex of G, as required. $\square$

To prove Theorem 1.3.2, we can now simply remove the vertex v given by Lemma 1.3.3, apply the induction hypothesis to find a bipartition of $G - v$, and then add v to one of the parts.

Second proof of Theorem 1.3.2 Let G be a minimal counterexample to the theorem, and observe that, as in the first proof of the theorem, we may assume that G is connected. By Lemma 1.3.3, there exists a vertex $v \in V(G)$ that is not a cut-vertex of G. Note that $G - v$ contains no odd cycle (since G does not), and is therefore bipartite, by the minimality of G. Let (A, B) be a bipartition of $G - v$; we claim that the neighbourhood $N_G(v)$ of v cannot intersect both A and B.

To see this, suppose (for a contradiction) that $a \in A$ and $b \in B$ are neighbours of v. Since v is not a cut-vertex of G, there exists a path P from a to b in $G - v$, and since a and b are in different parts of the bipartition of $G - v$, the length of P must be odd. Adding the edges av and bv to P, we obtain an odd cycle in G, contradicting our assumption.

It follows (without loss of generality) that v has no neighbours in A, and hence, setting $A' = A \cup \{v\}$, we obtain a partition $V(G) = A' \cup B$ into independent sets of G, as required. $\square$

It is often easier to work with bipartite graphs than general graphs. When it is, and we don't mind throwing away half of the edges (which will sometimes be the case), the following lemma is very useful.

Lemma 1.3.4. *Every graph G has a bipartite subgraph H with*

$$e(H) \geqslant \frac{e(G)}{2}.$$

We will give three proofs of Lemma 1.3.4: the first two provide algorithms for constructing H, while the third is non-constructive.

First proof of Lemma 1.3.4 Let H be a bipartite subgraph of G with the maximum number of edges. We claim that

$$d_H(v) \geqslant \frac{d_G(v)}{2} \tag{1.4}$$

for every $v \in V(G)$. Indeed, if (1.4) fails to hold for some vertex v, then we could construct a bipartite subgraph of G with more edges than H by moving v to the other part of the bipartition.

To be precise, let (A, B) be the bipartition of H, and suppose that $v \in A$ is the vertex for which (1.4) fails to hold. Then the bipartite graph $H' = G[A', B']$ with parts $A' = A \setminus \{v\}$ and $B' = B \cup \{v\}$ and edge set

$$E(H') = \big\{ ab \in E(G) : a \in A' \text{ and } b \in B' \big\}$$

satisfies

$$e(H') \geqslant e(H) - d_H(v) + \big(d_G(v) - d_H(v) \big) > e(H),$$

as claimed. By the handshaking lemma, it follows from (1.4) that

$$2e(H) = \sum_{v \in V(G)} d_H(v) \geqslant \sum_{v \in V(G)} \frac{d_G(v)}{2} = e(G),$$

as required. □

It is straightforward to modify the proof above to obtain a (fast) algorithm for constructing a bipartite subgraph with at least half the edges: starting with an arbitrary bipartition, repeatedly swap the part of any vertex for which (1.4) fails to hold (for the current bipartite subgraph H). Since the number of edges of H increases in each step, then the algorithm finds a suitable bipartition in at most $e(G)/2$ steps.

The following similar proof gives an even simpler algorithm.

Second proof of Lemma 1.3.4 Let $V(G) = \{v_1, \dots, v_n\}$, and recursively define sets A_k and B_k as follows. Starting from $A_0 = B_0 = \emptyset$, for each $k \in \{1, \dots, n\}$ add v_k to A_{k-1} if

$$|N_G(v_k) \cap A_{k-1}| \leqslant |N_G(v_k) \cap B_{k-1}|,$$

and add v_k to B_{k-1} otherwise. Writing H for the bipartite subgraph of G with bipartition (A_n, B_n), it follows that

$$e(H) \geqslant \frac{1}{2} \sum_{k=1}^{n} \big| N_G(v_k) \cap \{v_1, \dots, v_{k-1}\} \big| = \frac{e(G)}{2},$$

as required. □

Our third proof of Lemma 1.3.4 introduces a technique that will be a central theme of this book: the probabilistic method. As a warm-up, we first give the proof without using probabilistic language or notation.

Third proof of Lemma 1.3.4 Suppose, for a contradiction, that for every subset $A \subset V(G)$, the bipartite subgraph H_A of G with edge set

$$E(H_A) = \big\{ ab \in E(G) : a \in A \text{ and } b \in V(G) \setminus A \big\} \tag{1.5}$$

has fewer than $e(G)/2$ edges. We will count the pairs (e, A) such that $e \in E(H_A)$ in two ways. The key observation is that for every edge $e \in E(G)$, there are exactly $2^{v(G)-1}$ sets $A \subset V(G)$ such that $e \in E(H_A)$, since the only restriction is that exactly one of the vertices of e should be in A. It follows that

$$2^{v(G)-1} e(G) = \sum_{A \subset V(G)} e(H_A) < \sum_{A \subset V(G)} \frac{e(G)}{2} = 2^{v(G)-1} e(G),$$

giving the desired contradiction. □

We will now repeat the proof above in probabilistic language.

Third proof of Lemma 1.3.4, in other words We choose H randomly. To be precise, let $A \subset V(G)$ be chosen uniformly at random from all of the $2^{v(G)}$ subsets of $V(G)$, and let $H_A = G[A, A^c]$ be the induced bipartite subgraph of G defined in (1.5). Now, observe that each edge $uv \in E(G)$ is an edge of H_A with probability $1/2$, since the events

$$\big\{u, v \in A\big\}, \quad \big\{u, v \not\in A\big\}, \quad \big\{u \in A, v \not\in A\big\} \quad \text{and} \quad \big\{u \not\in A, v \in A\big\}$$

all have probability $1/4$. (We say that the events $\{u \in A\}$ and $\{v \in A\}$ are *independent*, meaning that the outcome of one of them does not affect the probability of the other.) It follows that the expected number of edges of H_A is

$$\mathbb{E}\big[e(H_A)\big] = \sum_{e \in E(G)} \mathbb{P}\big(e \in E(H_A)\big) = \frac{e(G)}{2},$$

and therefore there must exist a choice of A such that $e(H_A) \geqslant e(G)/2$. Since H_A is a bipartite subgraph of G, this proves the lemma. □

Before continuing, let's take a moment to recall a few basic concepts from probability theory that we used in the proof above:

- We will write $\mathbb{P}$ to denote a probability measure on a finite set Ω. (This just means that $\mathbb{P}$ assigns a probability in $[0, 1]$ to each element of Ω, and the probabilities add up to 1.) For example, in the proof above $\mathbb{P}$ assigns probability $2^{-v(G)}$ to each subset $A \subset V(G)$.
- A *random variable* is a function from Ω to $\mathbb{R}$; in the proof above we used the random variable $e(H_A)$. In this book our random variables will usually take values in the non-negative integers $\{0, 1, 2, \ldots\}$.
- The *expectation* of a random variable $X \colon \Omega \to \{0, 1, 2, \ldots\}$ is

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k \cdot \mathbb{P}(X = k). \tag{1.6}$$

We used two properties of the expectation in the proof above:

(i) Linearity: for *any* random variables X and Y, we have

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]. \tag{1.7}$$

(ii) A random variable cannot be always strictly less than its expectation. That is, for any random variable X we have

$$\mathbb{P}\big(X \geqslant \mathbb{E}[X]\big) > 0. \tag{1.8}$$

To be precise, we used linearity of expectation to calculate $\mathbb{E}\big[e(H_A)\big]$, together with the observations that

$$e(H_A) = \sum_{e \in E(G)} \mathbb{1}\big[e \in E(H_A)\big],$$

and that

$$\mathbb{E}\big[\mathbb{1}[e \in E(H_A)]\big] = \mathbb{P}\big(e \in E(H_A)\big),$$

and we used the fact that a random variable cannot be always strictly less than its expectation to deduce that there exists a set $A \subset V(G)$ such that $e(H_A) \geqslant \mathbb{E}\big[e(H_A)\big] = e(G)/2$.

To finish this section, let us quickly introduce another simple but important bipartite graph that we will encounter several times in later chapters: the *complete bipartite graph* $K_{s,t}$, defined by

$$V(K_{s,t}) = S \cup T \quad \text{and} \quad E(K_{s,t}) = \big\{uv : u \in S \text{ and } v \in T\big\},$$

where S and T are disjoint sets with $|S| = s$ and $|T| = t$. Note in particular that if G is a bipartite graph, then G is a subgraph of $K_{s,t}$ for some s and t. This observation will sometimes allow us to reduce the problem of proving a result for all bipartite graphs to proving it for the (much simpler) family of complete bipartite graphs.

We saw in this section that the property of being bipartite is quite simple: it just corresponds to avoiding odd cycles. In the next section we will study *r-partite graphs*; that is, graphs whose vertices can be partitioned into r independent sets. Perhaps surprisingly, when $r \geqslant 3$ it is *much* more difficult to determine whether a graph has this property.

1.4 The chromatic number of a graph

In the previous section we studied graphs whose vertices can be partitioned into two independent sets. In this section we will generalise this concept, allowing more than two sets in our partition. To do so, let us define an *r-colouring* of (the vertices of) a graph G to be a function

$$c : V(G) \to \{1, \ldots, r\},$$

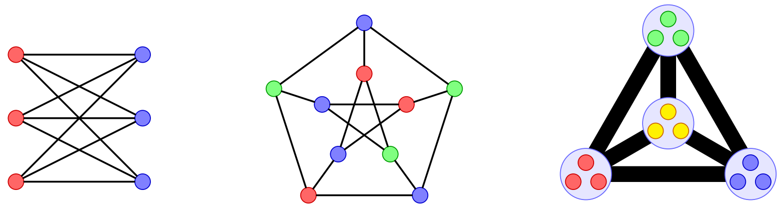

Figure 1.3 Graphs with chromatic number 2 (the complete bipartite graph $K_{3,3}$), chromatic number 3 (the Petersen graph), and chromatic number 4 (a complete 4-partite graph – each circle is an independent set, and the thick black lines indicate that all edges between the circles are included).

and say that a colouring c is *proper* if no edge of G is monochromatic, that is, if $c(u) \neq c(v)$ for every edge $uv \in E(G)$. We say that a graph G is *r-colourable* if there exists a proper r-colouring of G.

Definition 1.4.1. The *chromatic number* of a graph G, denoted by $\chi(G)$, is the smallest number r of colours for which G is r-colourable.

For example, observe that

$$\chi(K_k) = k, \qquad \chi(C_{2k}) = 2 \qquad \text{and} \qquad \chi(C_{2k+1}) = 3,$$

where C_k is the cycle of length k. Given a colouring c of the vertices of G, the *colour classes* of c are the sets

$$X_i = \big\{ v \in V(G) : c(v) = i \big\}$$

of vertices with the same colour. Note that the colour classes form a partition of $V(G)$, and that if c is a proper colouring then the colour classes are independent sets. In particular, a graph is bipartite if and only if there exists a proper 2-colouring of its vertices. More generally, it follows that being r-colourable is equivalent to being r-partite.

Observation 1.4.2. *A graph G is r-partite if and only if $\chi(G) \leqslant r$.*

Note that $\chi(G) \leqslant v(G)$ for every graph G, since we may colour each vertex with a different colour, and that equality holds if and only if G is a complete graph. For graphs with no high-degree vertices, a much better upper bound on $\chi(G)$ is given by the following simple theorem. Recall from (1.3) that we write $\Delta(G)$ to denote the maximum degree of a vertex in a graph G.

Theorem 1.4.3. *For every graph G,*

$$\chi(G) \leqslant \Delta(G) + 1.$$

Proof Let $r = \Delta(G) + 1$. We define a proper r-colouring

$$c : V(G) \to \{1, \dots, r\}$$

of G using the following 'greedy' algorithm: for each vertex in turn, choose any colour for v that has not yet been assigned to any of the neighbours of v. Since v has at most $\Delta(G)$ neighbours, and $r > \Delta(G)$, there is always an available choice. Since, for each edge $uv \in E(G)$, the colour of the second vertex to be coloured is always different to the first, it follows that c is a proper r-colouring of G, as required. □

Note that the bound given by Theorem 1.4.3 is sharp when $G = K_k$, since $\chi(K_k) = k$ and $\Delta(K_k) = k - 1$, and also when $G = C_{2k+1}$, since $\chi(C_{2k+1}) = 3$ and $\Delta(C_{2k+1}) = 2$. We will prove in Chapter 3 that these are the only connected graphs for which the bound is sharp.

When G has a small number of vertices of high degree, on the other hand, Theorem 1.4.3 can give a very bad bound. For example, if G is the *star* $K_{1,n-1}$ on n vertices, the theorem only gives the trivial bound $\chi(K_{1,n-1}) \leqslant n$, whereas it is easy to see that $\chi(K_{1,n-1}) = 2$. In this case, however, there is a very simple way to improve the bound given by the greedy algorithm: colour the high degree vertex first, and then colour the remaining vertices greedily.

Generalising this idea, we arrive at the following definition.

Definition 1.4.4. Let G be a graph, and let $d \in \mathbb{N}$. We say that G is *d-degenerate* if there exists an ordering $(v_1, \dots, v_n)$ of the vertices of G such that for each $i \in \{1, \dots, n\}$,

$$\big| N_G(v_i) \cap \{v_1, \dots, v_{i-1}\} \big| \leqslant d. \tag{1.9}$$

In other words, each vertex sends at most d edges 'backwards'.

We can now easily prove the following bound, which is sometimes much stronger than that given by Theorem 1.4.3.

Theorem 1.4.5. *If G is a d-degenerate graph, then*

$$\chi(G) \leqslant d + 1.$$

Proof We apply the greedy algorithm to the ordering $(v_1, \dots, v_n)$ of the vertices of G such that (1.9) holds for every $i \in \{1, \dots, n\}$. Since, for each vertex v_i, at most d of the neighbours of v_i are coloured before v_i, we obtain a proper $(d+1)$-colouring of G, as required. □

Theorem 1.4.5 has the following useful corollary.

Corollary 1.4.6. *A graph G contains every tree on $\chi(G)$ vertices.*

Proof Set $d = \chi(G) - 2$, and observe that G is not d-degenerate, by Theorem 1.4.5. Let $v_1, \ldots, v_k$ be a maximal sequence of distinct vertices of G with the property that

$$\big|N_G(v_i) \setminus \{v_1, \ldots, v_{i-1}\}\big| \leqslant d$$

for every $i \in \{1, \ldots, k\}$. Note that $k < n$, since G is not d-degenerate; indeed, if $k = n$ then the sequence $(v_n, \ldots, v_1)$ would satisfy (1.9). It follows that the graph $G[A]$ induced by the set $A = \{v_{k+1}, \ldots, v_n\}$ has minimum degree at least $d+1$, and hence, by Theorem 1.2.5, we have $T \subset G$ for every tree T with $d + 2 = \chi(G)$ vertices, as claimed. □

In the proof of Corollary 1.4.6, we used the fact that a graph that is not d-degenerate has a subgraph with minimum degree at least $d+1$. This turns out to be an equivalent definition of d-degeneracy.

Theorem 1.4.7. *Let G be a graph and let $d \in \mathbb{N}$. Then*

$$G \text{ is } d\text{-degenerate} \quad \Leftrightarrow \quad \delta(H) \leqslant d \text{ for every subgraph } H \subset G.$$

Proof Suppose first that $\delta(H) \leqslant d$ for every subgraph $H \subset G$. Then we can choose an ordering satisfying (1.9) greedily, simply by letting v_i be a vertex of degree at most d in the subgraph $H = G[A_i]$ induced by the set $A_i = V(G) \setminus \{v_{i+1}, \ldots, v_n\}$. On the other hand, if there exists a subgraph $H \subset G$ with $\delta(H) > d$, then there is no ordering of the vertices of G satisfying (1.9), since in any ordering the vertex of H with the largest index sends more than d edges backwards. □

How can we prove a lower bound on the chromatic number of a graph? One simple way to do so is to find a subgraph with large chromatic number, and then apply the following observation.

Observation 1.4.8. *If $H \subset G$, then $\chi(H) \leqslant \chi(G)$.*

Proof If $c : V(G) \to \{1, \ldots, r\}$ is a proper r-colouring of G, then restricting c to the vertices of H gives a proper r-colouring of H. □

In particular, in order to prove that the chromatic number of G is at least r, it is enough to find a clique on r vertices in G. In order to state this simple bound, let us write $\omega(G)$ for the *clique number* of G, that is, the number of vertices of the largest complete subgraph of G.

Observation 1.4.9. *For every graph G, we have $\chi(G) \geqslant \omega(G)$.*

The alert reader may have noticed that we actually already used Observation 1.4.8 in Section 1.3, to show that if G contains an odd cycle, then it is not bipartite.

Another simple way to bound the chromatic number from below is by bounding the size of the largest independent set in G from above. To state the next theorem, we define the *independence number* of G to be

$$\alpha(G) = \max\big\{|A| : A \subset V(G) \text{ is an independent set in } G\big\};$$

that is, the size of the largest set of vertices that contains no edge of G. The bound in the following theorem is an immediate consequence of the definitions of $\alpha(G)$ and $\chi(G)$.

Theorem 1.4.10. *Let G be a graph with n vertices. Then*

$$\chi(G) \geqslant \frac{n}{\alpha(G)}.$$

Proof Set $r = \chi(G)$, let

$$c : V(G) \to \{1, \ldots, r\}$$

be a proper r-colouring of G, and let $X_1, \ldots, X_r$ be the colour classes of c. Recall that each colour class is an independent set in G, and that the colour classes partition $V(G)$. It follows that

$$n \,=\, \sum_{i=1}^{r} |X_i| \,\leqslant\, r \cdot \alpha(G),$$

as claimed. □

Note that by combining Theorems 1.4.3 and 1.4.10, we obtain the following simple (but useful) bound on the independence number.

Corollary 1.4.11. *Let G be a graph with n vertices. Then*

$$\alpha(G) \geqslant \frac{n}{\Delta(G) + 1}.$$

It is not hard to construct graphs for which the bound on $\chi(G)$ given by Theorem 1.4.10 is very far from the truth. For example, any graph G containing both a clique and an independent set of size $n/2$ has chromatic number $n/2$, but Theorem 1.4.10 only implies that $\chi(G) \geqslant 2$. On the other hand, it turns out that the bound given by the theorem is not far from the truth for a *typical* graph. The proof of this beautiful and surprising fact is unfortunately beyond the scope of this book, but in Chapter 5 we will lay down some of the foundations that underlie it.

The lower bound in Observation 1.4.9 can also be far from sharp; indeed, we will show in Chapter 5 that there exist triangle-free graphs with arbitrarily large chromatic number.

1.5 Planar graphs, and the five colour theorem

Some of the oldest problems in graph theory are about *planar graphs*, that is, graphs that can be drawn in the plane without crossing edges. The most famous example is the 'four colour theorem', which was first conjectured in 1852 by Francis Guthrie, and states that the countries on any map can be coloured with four colours in such a way that neighbouring countries get different colours. In our language, this is equivalent to the statement that every planar graph has chromatic number at most 4. Unfortunately, the only known proofs of this theorem are very long and complicated; however, using the results and techniques that we have already met, we will be able to give a simple proof of the 'five colour theorem': that every planar graph has chromatic number at most five.

Let us begin by defining the family of planar graphs.

Definition 1.5.1. A graph G is *planar* if it can be drawn in $\mathbb{R}^2$ in such a way that no two edges of G cross each other.

To be slightly more precise, a graph G can be drawn in the plane without crossings if there exists an injective map $f\colon V(G) \to \mathbb{R}^2$ such that for every pair of distinct edges $uv, xy \in E(G)$ the line segments joining $f(u)$ to $f(v)$ and $f(x)$ to $f(y)$ only intersect (if at all) at their endpoints. Perhaps surprisingly, it turns out that allowing edges to be represented by either continuous curves or straight line segments defines the same family of graphs; this fundamental fact about planar graphs is known as Fáry's theorem, and is not at all easy to prove.

As was the case for trees, a natural first question is: how many edges can a planar graph with n vertices have? To construct a planar graph with 'many' edges, consider the following process. Starting with a triangle, repeatedly add a vertex inside a triangular face T, and add edges from this new vertex to the three vertices of T. By induction, any graph G constructed in this way is planar, and has exactly $3v(G) - 6$ edges.

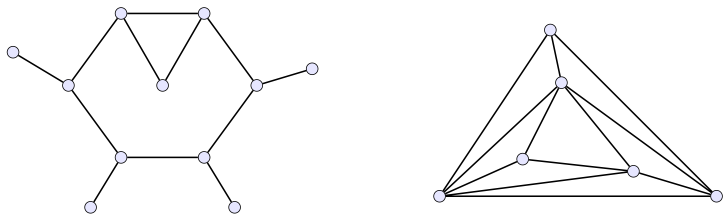

Figure 1.4 Two planar graphs: the first has 11 vertices, 12 edges and 3 faces; the second has 6 vertices, 12 edges and 8 faces.

The following theorem shows that this construction is best possible.

Theorem 1.5.2. *If G is a planar graph with $n \geqslant 3$ vertices, then*

$$e(G) \leqslant 3n - 6.$$

We will deduce Theorem 1.5.2 from the following famous formula of Euler, which was originally stated for convex polyhedra.

Euler's formula. *If G is a connected planar graph with n vertices, then*

$$n + f(G) = e(G) + 2,$$

where $f(G)$ is the number of faces in a drawing of G in the plane.

We remark that it is not obvious a priori that $f(G)$ is well-defined; however, this will follow from the proof of Euler's formula.

Proof of Euler's formula We will prove the claim by induction on $e(G)$. Note first that $e(G) \geqslant n - 1$, by Corollary 1.2.8, since G is connected. Moreover, if $e(G) = n - 1$ then G is a tree, by Theorem 1.2.10, and is therefore acyclic. It follows that $f(G) = 1$ (since for any graph G there is a unique unbounded face, and any bounded face is surrounded by a cycle), so in this case we have $n + 1 = e(G) + 2$, as required.

We may therefore assume that $e(G) \geqslant n$, and that Euler's formula holds for every connected planar graph with n vertices and at most $e(G) - 1$ edges. Since $e(G) \geqslant n$, it follows from Theorem 1.1.5 that G contains a cycle C. Let e be an edge of C, and let $G' = G - e$ be the graph obtained from G by deleting the edge e.

Note that G' is connected, by Lemma 1.2.7. Moreover, observe that $f(G') = f(G) - 1$, since removing the edge e from G allows the two faces on either side of e to merge, and leaves all other faces untouched.

It follows that G' is a connected planar graph with n vertices, and with $f(G) = f(G') + 1$. Therefore, applying the induction hypothesis to G', we deduce that

$$n + f(G) = n + f(G') + 1 = e(G') + 3 = e(G) + 2,$$

as required. □

We can now bound the number of edges in a planar graph.

Proof of Theorem 1.5.2 It will suffice to prove the theorem when G is connected, since we can then apply it to each component of G. Fix an embedding of G in the plane, and count the pairs (uv, f) where uv is a 'directed' edge of G (so uv is different from vu), and f is the face on the 'right-hand side' of the line segment from u to v. Note that each edge is counted twice (once for each orientation), and that each face is counted at least three times, since $n \geqslant 3$, and therefore every face has at least three (directed) edges on its boundary. It follows that

$$3 \cdot f(G) \leqslant 2 \cdot e(G).$$

Combining this with Euler's formula, we deduce that

$$e(G) + 2 = n + f(G) \leqslant n + \frac{2 \cdot e(G)}{3},$$

and hence $e(G) \leqslant 3n - 6$, as claimed. □

It follows immediately from Theorem 1.5.2 that every planar graph has a vertex of degree at most 5. Since we will use this fact below, let us record it as a corollary.

Corollary 1.5.3. *If G is a planar graph, then*

$$\delta(G) \leqslant 5.$$

Proof We may assume that $v(G) = n \geqslant 3$, since otherwise the claim holds trivially. Now, if $d_G(v) \geqslant 6$ for every $v \in V(G)$, then $e(G) \geqslant 3n$, by the handshaking lemma, contradicting Theorem 1.5.2. □

As mentioned above, the most famous fact about planar graphs is that their chromatic number is always at most 4. (Note that this bound is best possible, since the graph K_4 is planar.) The only known proofs of the four colour theorem are long and complicated, and involve reducing the theorem to a large number of special cases which can be checked by computer. We will therefore only prove the five colour theorem; that is, that every planar graph has chromatic number at most 5.

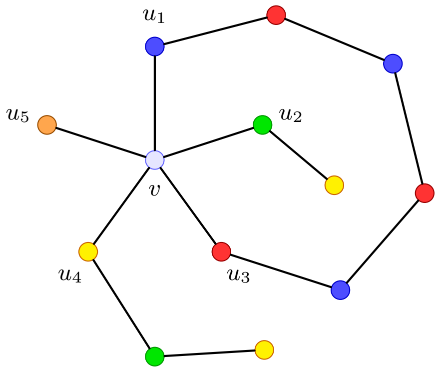

Figure 1.5 The first proof of the five colour theorem.

To get warmed up, let us first prove the (even weaker) six colour theorem, i.e., that $\chi(G) \leqslant 6$ for every planar graph G.

Theorem 1.5.4. *If G is a planar graph, then*

$$\chi(G) \leqslant 6.$$

Proof Let G be a minimal counterexample. By Corollary 1.5.3, there exists a vertex $v \in V(G)$ of degree at most 5. Note that the graph $G' = G - v$ is planar, and therefore, by the minimality of G, there exists a proper 6-colouring c of G'. Since v has only 5 neighbours, there exists a colour not used by any of those vertices, and we may therefore use this colour to extend c to a proper 6-colouring of G, as required. □

Proving the five colour theorem is not quite so easy.

Theorem 1.5.5. *If G is a planar graph, then*

$$\chi(G) \leqslant 5.$$

First proof of Theorem 1.5.5 Let G be a minimal counterexample. As in the proof of the six colour theorem, by Corollary 1.5.3 there exists a vertex $v \in V(G)$ with $d_G(v) \leqslant 5$, and by the minimality of G there exists a proper 5-colouring c of the graph $H = G - v$. Now, note that if there exists a colour that is not used by any of the neighbours of v, then we may use this colour to extend c to a proper 5-colouring of G. We may therefore assume that $d_G(v) = 5$, and that each of the neighbours of v receives a different colour in the colouring c.

The idea now is to recolour some of the vertices of H so that two of the neighbours of v receive the same colour, which will then allow us to colour v and obtain a proper 5-colouring of G. The recolouring procedure is as follows: let u_i be the neighbour of v with $c(u_i) = i$, let $j \neq i$ be a different colour, and let A_{ij} denote the set of vertices of H that can be reached from u_i by a path that only uses colours i and j. Observe that the following 5-colouring of H is proper:

$$c'(w) = \begin{cases} c(w) & \text{if} \quad w \notin A_{ij} \\ i & \text{if} \quad w \in A_{ij} \text{ and } c(w) = j \\ j & \text{if} \quad w \in A_{ij} \text{ and } c(w) = i; \end{cases}$$

that is, we swap the colours of the vertices in A_{ij}. Indeed, there is no monochromatic edge outside A_{ij} (since nothing changed), or inside A_{ij} (since each edge still has one vertex of colour i and one of colour j), or between A_{ij} and $V(H) \setminus A_{ij}$ (since the end-vertex not in A_{ij} cannot be coloured i or j, as otherwise it would have been included in A_{ij}).

Now, let u_j be the neighbour of v with $c(u_j) = j$, and suppose first that $u_j \notin A_{ij}$. Then c' is a proper colouring of H in which none of the neighbours of v is coloured i. We can therefore give colour i to v, and obtain a proper 5-colouring of G. Since i and j were arbitrary, we may therefore assume that $u_j \in A_{ij}$ for every $i, j \in \{1, \dots, 5\}$ with $i \neq j$.

To complete the proof, let us assume (without loss of generality) that the neighbours of v are cyclically ordered $u_1, \dots, u_5$, and consider the sets A_{13} and A_{24}. Observe that $H[A_{13}]$ contains a path from u_1 to u_3, and together with v this path forms a cycle C_{13} in G, with u_2 inside the cycle and u_4 outside it. Since G is planar, it follows that any path from u_2 to u_4 must intersect C_{13} in at least one vertex. But we assumed that $H[A_{24}]$ contains a path between u_2 and u_4, all of whose vertices are coloured either 2 or 4, while the vertices of A_{13} are all coloured either 1 or 3, so this is a contradiction. We may therefore recolour one of the sets A_{ij}, and obtain a proper 5-colouring of G, as required. $\square$

The proof above was discovered by Kempe in 1879, who claimed that it was in fact a proof of the four colour theorem! His mistake was only discovered 11 years later, by Heawood, who realised that it actually implied the bound given above.[3] The four colour theorem was finally proved almost a century later, by Appel and Haken, and was the first major theorem to be proved with the help of a computer.

[3] It is a worthwhile exercise to convince yourself that the proof of Theorem 1.5.5 cannot easily be adapted to prove the four colour theorem.

To finish this section, let us give a different proof of the five colour theorem, which uses an important concept: the 'contraction' of an edge. This is quite simple: contracting an edge $e = uv$ of a graph G means replacing the vertices u and v by a single vertex w, joining w to all the neighbours of u and v, and deleting any loops or multiple edges that arise in this process. That is, the contracted graph G/e has vertex set $(V(G) \setminus \{u, v\}) \cup \{w\}$, and the neighbourhood of w in G/e is the set

$$N_{G/e}(w) = \big(N_G(u) \cup N_G(v)\big) \setminus \{u, v\}.$$

For our purposes, the key observation is that if G is planar, then G/e is also planar. We can now give a second proof of the five colour theorem.

Second proof of Theorem 1.5.5 We use induction on $v(G)$. As in the first proof of the theorem, we may assume that $\delta(G) = 5$, since if there exists a vertex $v \in V(G)$ with $d_G(v) \leqslant 4$, then the proper 5-colouring of $G - v$ given by the induction hypothesis can be extended to a proper 5-colouring of G. Fix a vertex $x \in V(G)$ with $d_G(x) = 5$.

Now, since the graph K_5 is not planar (see Exercise 1.10), it follows that $K_5 \not\subset G$, and therefore there must be a pair of vertices $y, z \in N_G(x)$ with $yz \notin E(G)$. Let G' be the graph obtained from G by contracting the edges xy and xz, and observe that G' is planar and $v(G') < v(G)$, so there exists a proper 5-colouring c' of G', by the induction hypothesis. Using c', we can now define a proper 5-colouring c of $G - x$, by setting $c(u) = c'(u)$ for every $u \in V(G) \setminus \{x, y, z\}$ and $c(y) = c(z) = c'(w)$ (where w is the vertex formed by the contraction of xy and xz). Since two of the neighbours of x have the same colour in the colouring c, we can extend it to a proper 5-colouring of G, as required. $\square$

1.6 Triangle-free graphs

In Chapters 2 and 4, we will study the following general question:

"What can we say about an H-free graph?"

In the next two sections we will warm ourselves up for those chapters by considering some special cases of this problem. We will begin here by considering the case in which H is the 'triangle' K_3.

We will say that graph is *triangle-free* if it does not contain a copy of K_3. The family of triangle-free graphs is often useful as a relatively simple example of more general and complex behaviour.

Let G be a triangle-free graph with n vertices. Once again, a natural first question is: how many edges can G have? Since every subgraph of a triangle-free graph is also triangle-free, we are really asking for the *maximum* number of edges in a triangle-free graph; that is, for the *extremal number* of the triangle,

$$\mathrm{ex}(n, K_3) = \max\big\{e(G) : K_3 \not\subset G \subset K_n\big\}.$$

Note that every bipartite graph is triangle-free, so a lower bound on $\mathrm{ex}(n, K_3)$ is given by the bipartite graph with most edges; that is, by the complete bipartite graph with part sizes as equal as possible. It is easy to check that this graph has $\lfloor n^2/4 \rfloor$ edges; the following theorem shows that it is extremal for K_3.

Theorem 1.6.1. *If G is a triangle-free graph with n vertices, then*

$$e(G) \leqslant \frac{n^2}{4}.$$

We will give two proofs of Theorem 1.6.1; both are very simple. For the first, recall that we write $\Delta(G)$ for the maximum degree of G.

First proof of Theorem 1.6.1 Let $u \in V(G)$ be a vertex of maximum degree in G. Observe that $N_G(u)$ is an independent set (since G is triangle-free), and therefore

$$d_G(v) \leqslant n - d_G(u) = n - \Delta(G)$$

for every $v \in N_G(u)$. By the handshaking lemma, it follows that

$$\begin{aligned} e(G) &\leqslant \frac{1}{2}\bigg(\sum_{v \in N_G(u)} \big(n - \Delta(G)\big) + \sum_{v \in V(G) \setminus N_G(u)} \Delta(G) \bigg) \\ &= \Delta(G)\big(n - \Delta(G)\big) \leqslant \frac{n^2}{4}, \end{aligned}$$

since the function $x(n-x)$ is maximised by taking $x = n/2$. □

Our second proof uses induction on the number of vertices.

Second proof of Theorem 1.6.1 Observe first that the statement holds vacuously if $n = 1$ or $n = 2$, since in those cases there does not exist any graph with n vertices and more than $n^2/4$ edges. So let $n \geqslant 3$, and assume that the claim holds for every graph with $n-2$ vertices.

Let G be a triangle-free graph with n vertices; we are required to show that G has at most $n^2/4$ edges. Let uv be an edge of G, and let $H = G - u - v$ be the graph obtained from G by removing the vertices

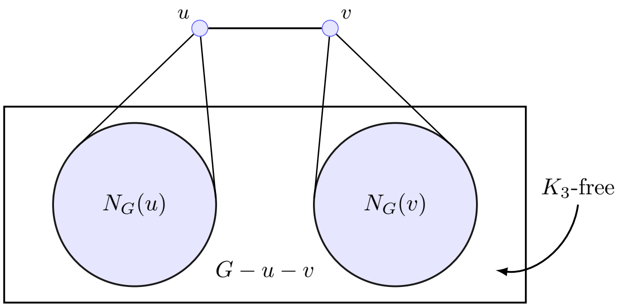

Figure 1.6 The second proof of Theorem 1.6.1.

u and v (and all edges incident to either). Since H is triangle-free and has $n-2$ vertices, by the induction hypothesis we have

$$e(H) \leqslant \frac{(n-2)^2}{4}.$$

Now, observe that each vertex of H has at most one neighbour in the set $\{u, v\}$, since uv is an edge and G is triangle-free. It follows that

$$e(G) \;\leqslant\; e(H) + n - 1 \;\leqslant\; \frac{(n-2)^2}{4} + n - 1 \;=\; \frac{n^2}{4},$$

as required. □

It is not hard to modify the proofs above to show that the complete bipartite graph $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$ is the unique *extremal graph* for K_3, that is, the unique triangle-free graph on n vertices with the maximum possible number of edges. For example, if G is triangle-free and $e(G) = \lfloor n^2/4 \rfloor$, then in the first proof we must have $\Delta(G) = \lceil n/2 \rceil$, and every vertex of $N_G(u)$ must be connected to every vertex of $V(G) \setminus N_G(u)$.

What is the densest triangle-free graph that is not bipartite? When answering this question we would like to forbid uninteresting examples such as the disjoint union of a complete bipartite graph and a copy of C_5, so instead of asking to maximise the number of edges, let us instead use the minimum degree as our measure of density.

Question 1.6.2. How large is

$$\max\big\{\delta(G) : K_3 \not\subset G \subset K_n \text{ and } \chi(G) \geqslant 3\big\}?$$

A simple way to construct triangle-free graphs with large minimum degree is to 'blow up' small triangle-free graphs, replacing each vertex by a large set of vertices and each edge by a complete bipartite graph.

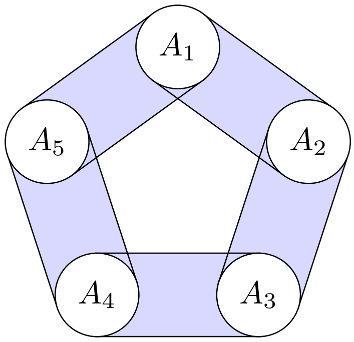

Figure 1.7 A blow-up of C_5; each of the sets A_i is independent, and the blue connectors indicate that all edges between the two sets are included.

Definition 1.6.3. Let G be a graph with vertex set $\{v_1, \ldots, v_n\}$, and let $t \in \mathbb{N}$. The *t-blow-up* of G, denoted $G(t)$, is the graph with vertex set

$$V\big(G(t)\big) = A_1 \cup \cdots \cup A_n,$$

where $A_1, \ldots, A_n$ are disjoint sets of size t, and edge set

$$E\big(G(t)\big) = \bigcup_{v_i v_j \in E(G)} \big\{xy : x \in A_i \text{ and } y \in A_j\big\}.$$

Note that if G is triangle-free, then $G(t)$ is also triangle-free, since there are no edges inside the sets A_i. Observe also that

$$\delta\big(G(t)\big) = t \cdot \delta(G) \qquad \text{and} \qquad \chi\big(G(t)\big) = \chi(G).$$

In particular, the graph $C_5(t)$ (the t-blow-up of C_5) is triangle-free, is not bipartite, and has $5t$ vertices and minimum degree $2t$. The following theorem, which answered a question of Erdős, was proved by Andrásfai.

Theorem 1.6.4. *If G is a triangle-free graph with n vertices and*

$$\delta(G) > \frac{2n}{5},$$

then G is bipartite.

In the proof of Theorem 1.6.4, the first step will be to find three vertices u, v and w such that $uv \in E(G)$ and $uw, vw \notin E(G)$. The following simple lemma will allow us to do so. A *complete r-partite graph* is a graph G for which there exists a partition $V(G) = A_1 \cup \cdots \cup A_r$ of the vertices with the following property: $uv \in E(G)$ if and only if u and v are in different parts of the partition.

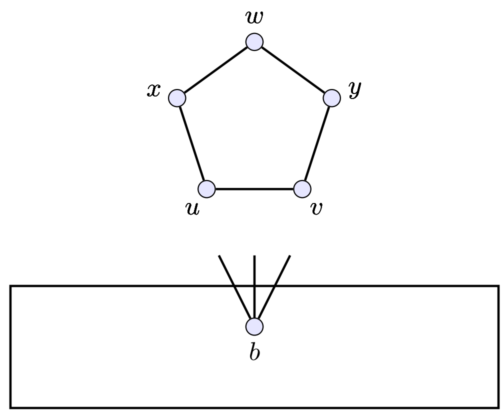

Figure 1.8 The first proof of Andrásfai's theorem.

Lemma 1.6.5. *Let G be a graph, and suppose that there do not exist $u, v, w \in V(G)$ such that*

$$uv \in E(G) \qquad \text{and} \qquad uw, vw \notin E(G). \tag{1.10}$$

Then G is a complete r-partite graph for some $r \in \mathbb{N}$.

Proof We simply apply Theorem 1.1.7 to G^c, the complement of G. Indeed, if no three vertices $u, v, w \in V(G)$ satisfy (1.10), then G^c is induced-P_2-free, and G is a complete r-partite graph for some $r \in \mathbb{N}$ if and only if G^c is a union of vertex-disjoint cliques. □

We can now prove Andrásfai's theorem.

First proof of Theorem 1.6.4 Let G be a triangle-free graph with n vertices and $\delta(G) > 2n/5$, and suppose that G is not bipartite. We may assume (by adding edges if necessary) that G is maximal triangle-free, meaning that the addition of any new edge to G would create a triangle. Note that G is not a complete r-partite graph, since it is triangle-free and not bipartite, so by Lemma 1.6.5 there exist vertices $u, v, w \in V(G)$ such that $uv \in E(G)$ and $uw, vw \notin E(G)$.

Now, by the maximality of G, there exist vertices $x, y \in V(G)$ such that $x \in N_G(u) \cap N_G(w)$ and $y \in N_G(v) \cap N_G(w)$. Moreover, since G is triangle-free, we have $x \neq y$ (otherwise $\{x, u, v\}$ would be the vertex set of a triangle). Set $A = \{u, v, w, x, y\}$, and observe that (w, x, u, v, y, w) is an induced copy of C_5 in $G[A]$, since $wx, xu, uv, vy, yw \in E(G)$, and any other edge inside A would complete a triangle.

Now, set $B = V(G) \setminus A$, and observe that

$$e\big(G[A,B]\big) \geqslant 5\big(\delta(G) - 2\big) > 2(n-5),$$

where $G[A,B]$ is the bipartite graph with parts A and B and edge set $\{ab \in E(G) : a \in A,\, b \in B\}$. By the pigeonhole principle, it follows that there exists a vertex $b \in B$ with at least three neighbours in A. But $G[A] = C_5$, and C_5 does not have an independent set of size three, so this implies that G contains a triangle, as required. □

Our second proof uses Theorem 1.3.2 to find an odd induced cycle.

Second proof of Theorem 1.6.4 If G is not bipartite, then it contains an odd cycle, by Theorem 1.3.2. Let $C = (v_1, \dots, v_{2k+1}, v_1)$ be an odd cycle of minimum length in G, and observe that C is an induced cycle, since any edge $v_i v_j \in E(G) \setminus E(C)$ would give a shorter odd cycle.

Now the crucial observation: if G is triangle-free, then every vertex of G has at most two neighbours in $V(C)$. To see this, suppose that $u \in V(G) \setminus V(C)$ has three neighbours $x, y, z \in V(C)$, and partition the edges of C into three paths P_1, P_2 and P_3 with end-vertices x, y and z. Note that at least one of these paths has odd length; by the minimality of C, this path has to have length $2k-1$. This implies that the other two paths have length 1, contradicting our assumption.

We can now double-count the edges of G between the sets $A = V(C)$ and $B = V(G) \setminus A$, as in the first proof. We obtain

$$(2k+1)\big(\delta(G) - 2\big) \leqslant e\big(G[A,B]\big) \leqslant 2\big(n - 2k - 1\big),$$

and hence $\delta(G) \leqslant 2n/(2k+1) \leqslant 2n/5$, as required. □

To finish this section, we will prove one more simple but important property of triangle-free graphs.

Theorem 1.6.6. *If G is a triangle-free graph with n vertices, then*

$$\alpha(G) \geqslant \sqrt{n} - 1.$$

The theorem is an easy consequence of Corollary 1.4.11.

Proof of Theorem 1.6.6 Observe first that $\alpha(G) \geqslant \Delta(G)$, since the neighbourhood of every vertex is an independent set. We may therefore assume that $\Delta(G) < \sqrt{n} - 1$, which implies that

$$\alpha(G) \geqslant \frac{n}{\Delta(G) + 1} > \sqrt{n}$$

by Corollary 1.4.11, as required. □

The graph C_5 shows that we cannot always find an independent set of size $\sqrt{n}$ in a triangle-free graph; however, it turns out that when n is large, one can find much larger independent sets in triangle-free graphs with n vertices. We will return to this question in Chapter 4.

1.7 Ramsey theory

We showed in Theorem 1.6.6 that a triangle-free graph with n vertices has an independent set of size $\sqrt{n} - 1$. Another way of saying the same thing is that if $n > k^2$ and we colour each edge of K_n either red or blue, then there is either a red triangle, or a blue clique of size k.

What is the smallest value of n such that the same conclusion holds? This turns out to be a difficult question, so let us begin by considering an easier one: the case $k = 3$. That is, what is the smallest $n \in \mathbb{N}$ such that every red-blue colouring of the edges of K_n contains a monochromatic triangle (i.e., a triangle all of whose edges are the same colour)?

By Theorem 1.6.6, it suffices to take $n \geqslant 10$. On the other hand, it is not too hard to find a colouring of the edges of K_5 that contains no monochromatic triangle (find it!), so taking $n = 5$ is not enough. It turns out that this construction is best possible, and six vertices suffice.

Theorem 1.7.1. *If we colour each edge of K_6 either red or blue, then there is always a monochromatic triangle.*

Proof Let v be one of the vertices of K_6, and consider the colours of the five edges leaving v. By the pigeonhole principle, at least three of them must be the same colour. Assume (without loss of generality) that this colour is red, and let vx, vy and vz be red edges.

Now, if any of the edges xy, xz and yz are red, then we have a red triangle. But if not, then xyz is a blue triangle, so we are done. $\square$

Ramsey theory is the study of questions of this type. In particular, the *Ramsey numbers* are defined as follows.

Definition 1.7.2. For each $k \in \mathbb{N}$, the Ramsey number $R(k)$ is the minimum $n \in \mathbb{N}$ such that every red-blue colouring of the edges of K_n contains a monochromatic copy of K_k.

For example, Theorem 1.7.1 states that $R(3) \leqslant 6$, and partitioning the edges of K_5 into two copies of C_5 shows that $R(3) \geqslant 6$, so we have determined $R(3)$. The following theorem, proved by Ramsey in 1930, shows that the Ramsey number $R(k)$ exists for every $k \in \mathbb{N}$.

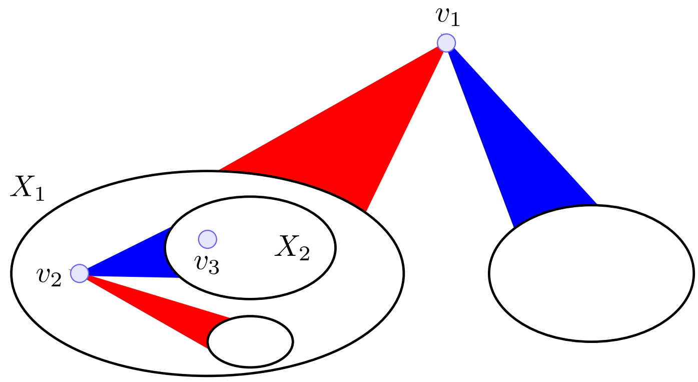

Figure 1.9 The proof of Ramsey's theorem.

Theorem 1.7.3. *For every $k \in \mathbb{N}$, there exists $n \in \mathbb{N}$ such that every red-blue colouring of $E(K_n)$ contains a monochromatic copy of K_k.*

Proof Let c be a red-blue colouring of the edges of K_n, let v_1 be an arbitrary vertex of K_n, and observe that, by the pigeonhole principle, there exists a colour a_1 such that the set

$$X_1 = \big\{ u \in V(K_n) : c(uv_1) = a_1 \big\}$$

contains at least $(n-1)/2$ vertices.

Now choose an arbitrary vertex $v_2 \in X_1$, and repeat the process. By the pigeonhole principle, there exists a colour a_2 and a set

$$X_2 = \big\{ u \in X_1 : c(uv_2) = a_2 \big\}$$

of size at least $(|X_1| - 1)/2$. Continuing in the same way, we obtain a sequence of sets

$$V(K_n) = X_0 \supset X_1 \supset \cdots \supset X_t \qquad \text{with} \qquad |X_i| \geqslant \frac{n}{2^i} - 1,$$

and also vertices $v_1, \dots, v_t$ and colours $a_1, \dots, a_t$ such that

$$v_i \in X_{i-1} \qquad \text{and} \qquad X_i = \big\{ u \in X_{i-1} : c(uv_i) = a_i \big\}$$

for every $1 \leqslant i \leqslant t$. Crucially, observe that

$$c(v_i v_j) = a_i \qquad \text{for every} \qquad 1 \leqslant i < j \leqslant t, \tag{1.11}$$

since $v_j \in X_{j-1} \subset X_i$. We can continue this process until we run out of vertices; that is, until $X_t = \emptyset$. Note that if n is sufficiently large, then by our lower bound on $|X_i|$, this does not occur until $t \geqslant 2k$.

Now, again using the pigeonhole principle, there exists a colour a (either red or blue) which appears at least k times in the sequence $(a_1, \ldots, a_{2k-1})$. Since $c(v_i v_j) = a$ whenever $a_i = a$, by (1.11), it follows that the set of vertices v_i such that $a_i = a$ induces a monochromatic clique (in colour a) of size at least k, as required. □

The proof of Ramsey's theorem given above shows that $R(k) \leqslant 4^k$. How close is this to the truth? To get a feel for this problem, let's begin by proving the following simple lower bound on $R(k)$.

Observation 1.7.4. *For every $k \in \mathbb{N}$, we have*

$$R(k) > (k-1)^2.$$

Proof Let $n = (k-1)^2$, and consider the following colouring of the edges of K_n. The red edges form $k-1$ disjoint copies of K_{k-1}, and the remaining edges are blue. Note that the blue edges form a complete $(k-1)$-partite graph, and therefore do not contain a copy of K_k (by Observation 1.4.9). The red edges also do not contain a copy of K_k, so this proves that $R(k) > (k-1)^2$, as claimed. □

Note that in the case $k = 3$ this construction is not best possible, since it only shows that $R(3) \geqslant 5$, and we saw above that $R(3) = 6$. For larger values of k, however, it turns out to be very difficult to construct a colouring with more than $(k-1)^2$ vertices and no monochromatic K_k (as a challenge, try to do so for $k = 4$; see also Exercise 1.31).

Nevertheless, in 1947 Erdős found a shockingly simple (and revolutionary) *probabilistic* proof of an exponential lower bound on $R(k)$.

Theorem 1.7.5.

$$R(k) \geqslant 2^{k/2}$$

for all $k \geqslant 5$.

As we did in Section 1.3 for our first probabilistic proof, we will give the proof of Theorem 1.7.5 twice, first avoiding probabilistic language, and then using it. In both cases we will need the following bounds on the binomial coefficient, which we will use several times in this book:

$$\left(\frac{n}{k}\right)^k \leqslant \binom{n}{k} \leqslant \left(\frac{en}{k}\right)^k \tag{1.12}$$

for every $1 \leqslant k \leqslant n$.

Proof of Theorem 1.7.5 Observe that there are $2^{\binom{n}{2}}$ colourings of the edges of K_n with two colours. The idea is to count the colourings of $E(K_n)$ with a monochromatic copy of K_k. If we can show that there are fewer than $2^{\binom{n}{2}}$ such 'bad' colourings, then it follows that there must exist a 'good' colouring, i.e., with no monochromatic K_k.

To bound the number of bad colourings, note that the number of ways to colour the edges of K_n red and blue such that a fixed copy of K_k is monochromatic is exactly $2^{\binom{n}{2}-\binom{k}{2}+1}$. Summing over choices of this K_k (that is, using the 'union bound'), it follows that there are at most

$$\binom{n}{k} 2^{\binom{n}{2}-\binom{k}{2}+1}$$

bad colourings. Now, if $n < 2^{k/2}$, then by (1.12) this is at most

$$2^{\binom{n}{2}+1}\left(\frac{en}{k}\cdot 2^{-(k-1)/2}\right)^k < 2^{\binom{n}{2}+1}\left(\frac{e\sqrt{2}}{k}\right)^k. \tag{1.13}$$

Finally, if $k \geqslant 5$ then the right-hand side of (1.13) is at most $2^{\binom{n}{2}}$, as required. It follows that there are more colourings than bad colourings, and therefore there must exist a good colouring, as claimed. □

In probabilistic language this proof can be described more succinctly: we simply colour randomly! The calculation that shows this works is almost identical to that above, but instead of counting colourings, we count the expected number of monochromatic cliques of size k.

Proof of Theorem 1.7.5, in other words Consider a random colouring of the edges of the complete graph K_n, chosen as follows: each edge is coloured red with probability 1/2, independently of all other edges, and is coloured blue otherwise. (Note that every red-blue colouring is chosen with the same probability, so this is the same as choosing the colouring uniformly at random.) Let X be the number of monochromatic copies of K_k in this colouring. We claim that, by linearity of expectation,

$$\mathbb{E}[X] = \binom{n}{k} 2^{-\binom{k}{2}+1}.$$

To see this, observe that there are $\binom{n}{k}$ copies of K_k in K_n, and each is monochromatic with probability exactly $2^{-\binom{k}{2}+1}$. Hence, by (1.12),

$$\mathbb{E}[X] \leqslant 2\cdot\left(\frac{en}{k}\cdot 2^{-(k-1)/2}\right)^k < 2\cdot\left(\frac{e\sqrt{2}}{k}\right)^k < 1$$

if $n < 2^{k/2}$ and $k \geqslant 5$. Since X takes values in the non-negative integers, it follows that $\mathbb{P}(X = 0) > 0$, and therefore there exists a colouring with $X = 0$ (that is, with no monochromatic copy of K_k), as required. □

We will discuss Ramsey numbers (and many other related questions) in much more detail in Chapter 4.

1.8 Exercises

Exercise 1.1. Show that every tree with at least two vertices has at least two leaves.

Exercise 1.2. A *forest* is a graph each of whose connected components is a tree. Show that

$$G \text{ is a forest} \quad \Leftrightarrow \quad G \text{ is acyclic} \quad \Leftrightarrow \quad G \text{ is 1-degenerate.}$$

Deduce (without using Theorem 1.3.1) that forests are bipartite.

Exercise 1.3. Let G be a forest with n vertices and k components. Show that G has exactly $n - k$ edges.

Exercise 1.4. Let F and H be two forests with the same vertex set. Show that if $e(F) < e(H)$, then there exists an edge of H that can be added to F without creating a cycle.

Exercise 1.5. Let F be a forest. Show that there exists a tree T with vertex set $V(F)$ such that $F \subset T$. Show moreover that if $F \subset G$ for some connected graph G, then G has a spanning tree T with $F \subset T$.

Exercise 1.6. Show that if G is a graph with $\Delta(G) \leqslant 2$, then every component of G is either a path or a cycle.

Exercise 1.7. Show that if G is not a forest, then it contains a cycle of length at least $\delta(G) + 1$.

Exercise 1.8. Prove that if the degree of every vertex in a graph G is even, then the edges of G can be partitioned into cycles.

Exercise 1.9. Show that if G is a connected graph and $v \in V(G)$, then there exists a walk of length at most $2n - 2$, starting and ending at v, that passes through every vertex of G.

Exercise 1.10. Prove that the graphs K_5 and $K_{3,3}$ are not planar.

Exercise 1.11. Let G be a triangle-free planar graph with n vertices. Show that if $n \geqslant 3$, then G has at most $2n - 4$ edges.

Exercise 1.12. Give an alternative proof of the six colour theorem, that $\chi(G) \leqslant 6$ for every planar graph G, using Theorem 1.4.5.

Exercise 1.13. Let G be a planar graph with no cycles of length 3, 4 or 5. Show that $\chi(G) \leqslant 3$.

Exercise 1.14. Prove that

$$e(G) \geqslant \binom{\chi(G)}{2}$$

for every graph G.

Exercise 1.15. Show that, for every $d \in \mathbb{N}$, there exist graphs with average degree arbitrarily close to $2d$, with no subgraph of minimum degree greater than d.

Exercise 1.16. Let G be a graph with at least one edge. Show that G has a bipartite subgraph H with

$$e(H) > \frac{e(G)}{2}.$$

Exercise 1.17. Let G be a graph in which k vertices have odd degree. Show that G has a bipartite subgraph H with

$$e(H) \geqslant \frac{e(G)}{2} + \frac{k}{4}.$$

Exercise 1.18. Show that every graph G has a subgraph H with

$$e(H) \geqslant \left(1 - \frac{1}{r}\right)e(G) \qquad \text{and} \qquad \chi(H) \leqslant r.$$

Exercise 1.19. Let G be a graph with n vertices and $e(G) \leqslant p\binom{n}{2}$. Show that, for every $m \leqslant n$, there exists an induced subgraph $H \subset G$ with m vertices and at most $p\binom{m}{2}$ edges.

Exercise 1.20. A graph is d-regular if every vertex has degree d. Show that if G is a d-regular graph with n vertices, then $\alpha(G) \leqslant n/2$.

Exercise 1.21. Let G be a graph with $\delta(G) \geqslant d$ and

$$v(G) \leqslant d \sum_{t=0}^{k-1} (d-1)^t. \tag{1.14}$$

Show that G contains a cycle of length at most $2k$.

Exercise 1.22. Construct a 3-regular graph on 10 vertices that has no copy of K_3 or C_4 (and find a nice way to draw it). Deduce that (1.14) is tight when $d = 3$ and $k = 2$. Is it also tight when $d = 4$ and $k = 2$?

Exercise 1.23. Show that the graph G that you constructed in the previous exercise (which is known as the Petersen graph) is the unique graph with 10 vertices, 15 edges and no copy of K_3 or C_4.

Exercise 1.24. Show that there is a bijection between the vertices of the Petersen graph and the subsets of size 2 of a set of size 5 such that edges map to disjoint pairs of sets.

Exercise 1.25. Show that the Petersen graph is not planar. If we remove an edge, do we obtain a planar graph?

Exercise 1.26. Let G be a graph on n vertices. Show that if G does not have a cycle of length at most $2\log_2 n$, then $\chi(G) \leqslant 3$.

Exercise 1.27. Let G be a connected graph. Show that

$$\chi(G - v) \leqslant \Delta(G)$$

for every $v \in V(G)$.

Exercise 1.28. Let G be a graph, and suppose that $\chi(G - v) \leqslant k$ for every $v \in V(G)$. Show that if G has a cut-vertex, then $\chi(G) \leqslant k$.

Exercise 1.29. Show that if $E(G) = E(H_1) \cup E(H_2)$, then

$$\chi(G) \leqslant \chi(H_1) \cdot \chi(H_2).$$

Deduce that

$$2\sqrt{n} \leqslant \chi(G) + \chi(G^c) \leqslant n + 1$$

for every graph G with n vertices. [Hint: use Theorems 1.4.5 and 1.4.7.]

Exercise 1.30. Show that every red-blue colouring of the edges of K_6 contains a monochromatic copy of C_4.

Exercise 1.31. Show that $R(4) \leqslant 18$.

Exercise 1.32. Show that if $n \geqslant 6$, then

$$\alpha(G) \geqslant \sqrt{n}$$

for every triangle-free graph G with n vertices.

Exercise 1.33. Show that if G is a triangle-free graph with $2k + 1$ vertices, then G^c contains a path of length k.

Exercise 1.34. Show that if G is a triangle-free graph with $2k+1$ vertices, then G^c contains every tree with k edges.

Exercise 1.35. Show that every red-blue-green colouring of $E(K_n)$ contains a monochromatic connected graph with at least $n/2$ vertices.

Exercise 1.36. Let G be a graph that contains every tree T with n vertices as a subgraph. Show that

$$e(G) \geqslant \frac{n \log n}{2}.$$

Exercise 1.37. Let T be a tree, and let $T_1, \dots, T_k \subset T$ be trees that pairwise intersect. Show that there is a vertex of T that is contained in all of the trees T_i.

Exercise 1.38. Prove that if G is a tree, then the vertices of G can be partitioned into $\alpha(G)$ edges and vertices.

Exercise 1.39. Let T be a tree with n vertices and at most ℓ leaves. Show that T contains at least

$$\frac{n}{3} - \ell$$

vertex-disjoint paths of length 2.

Exercise 1.40. Let G be a bipartite graph with parts A and B, where $|A| = |B| = n$. If $d_G(u) \geqslant a$ for every $u \in A$, and $d_G(v) \geqslant b$ for every $v \in B$, then G contains a matching with $\min\{n, a+b\}$ edges.

Exercise 1.41. Let G be a graph with n vertices and m edges, and suppose that G is drawn in the plane, but with edges allowed to cross.

(a) Show that at least $m - 3n + 6$ pairs of edges cross.

(b) By considering a random subset of the vertices, show that if $m \geqslant 4n$, then at least $m^3/64n^2$ pairs of edges cross.

Exercise 1.42. Given a tree with vertex set $\{1, \dots, n\}$, construct a sequence $(a_1, \dots, a_{n-2}) \in \{1, \dots, n\}^{n-2}$ in the following way: repeatedly remove the 'smallest' leaf u, and write down the neighbour of u.

Show that this process defines a bijection, and deduce that the complete graph K_n has exactly n^{n-2} different spanning trees.

Exercise 1.43. Show that for every graph G, there exists a partition $V(G) = A \cup B$ such that all degrees in $G[A]$ and $G[B]$ are even.

Historical notes

Euler's formula is one of the oldest results in graph theory; it was first observed for the Platonic solids by Francesco Maurolico in 1537, stated for general convex 3-dimensional polyhedra in 1758 by Euler (who gave an incorrect proof), and was finally proved in 1794 by Legendre, using spherical angles. However, it was not until 1813 that it was noticed, by Cauchy, that the formula holds in the more general setting of connected planar graphs. A closely-related theorem was proved by Descartes in the 1630s, but was only discovered over 200 years later (the story of the manuscript's survival, including three days at the bottom of the Seine following a shipwreck, is quite extraordinary). Euler's formula is a special case of the Euler characteristic (sometimes called the Euler–Poincaré characteristic), an invariant used in algebraic topology.

The study of the chromatic number of planar graphs was initiated relatively recently: the four-colour conjecture was first mentioned in a letter from De Morgan to Hamilton in 1852, and was first published as an open problem in the magazine *The Athenaeum* two years later. De Morgan had been told of the conjecture by his student Frederick Guthrie, who had learnt of it from his brother, Francis Guthrie (see Wilson's book for more on the origins of the problem). Kempe published a 'proof' of the conjecture in 1879, but Heawood showed 11 years later that his proof was incorrect, and modified it to prove the five colour theorem (this is the proof that we presented in Section 1.5). Another attempted proof, published by Tait in 1880, was shown to be incorrect by Petersen in 1898; his counterexample to Tait's approach was the graph in Exercise 1.22. The four colour theorem was finally proved in 1976 by Appel and Haken, with the help of a computer program written by John Koch, and using a new technique, pioneered by Heinrich Heesch in the 1960s, which allowed the problem to be reduced to checking (by computer) a large number of small cases. A simpler (but still very complicated) version of Appel and Haken's proof was found by Robertson, Sanders, Seymour and Thomas in 1997, and a formal (computer-checkable) version of the proof (using the Coq proof assistant) was published in 2005.

The study of trees (in particular, spanning trees) can be traced back to 1847, and the work of Kirchhoff on electrical networks. Another important early paper, to which the name 'graph' can traced, was written by Petersen in 1891, and many of the basic facts about graphs that we proved in this chapter were first collected in the classic 1936 textbook of Kőnig, who also proved many of the fundamental results of the area; we will see several examples of this in Chapter 3.

As we will see in Chapter 2, the study of the extremal number of a graph was initiated in 1941 by Pál Turán, who determined $\mathrm{ex}(n, K_r)$ exactly for every $n \geqslant r \geqslant 3$. The case $r = 3$ of Turán's theorem, which we proved in Section 1.6, is commonly referred to as Mantel's theorem. The first author is guilty of spreading around this incorrect attribution. In 1907, Mantel published this statement in the problem section of the journal *Wiskundige Opgaven*, but the solution by W. A. Wythoff published in a later issue of the journal was incorrect. When Turán proved his theorem, he was unaware of Mantel's problem.

Ramsey's theorem was first proved in 1930, and rediscovered a few years later by Erdős and Szekeres, whose proof we saw in Section 1.7. Over the next several decades, Erdős was the driving force behind the study of Ramsey theory, introducing (and proving fundamental results about) many of the variants that we will study in Chapter 4; in particular, he proved his famous exponential lower bound on $R(k)$ in 1947.

Finally, we would like to acknowledge that the introduction to graph theory that we have given in this chapter may have been too fast-paced for many readers. Those readers who would prefer a gentler introduction to graph theory can find it in the books listed below.

Further reading

A. Benjamin, G. Chartrand and P. Zhang, The Fascinating World of Graph Theory, Princeton University Press, 2017.

G. Chartrand, L. Lesniak and P. Zhang, Graphs & Digraphs (6th edition), CRC Press, 2016.

G. Chartrand and P. Zhang, A First Course in Graph Theory, Dover, 2012.

L. Lovász, J. Pelikán and K. Vesztergombi, Discrete Mathematics: Elementary and Beyond, Springer, 2003.

D. B. West, Introduction to Graph Theory (2nd edition), Pearson, 2018.

R. Wilson, Four Colors Suffice: How the Map Problem Was Solved, Princeton University Press, 2014.